

# Thanh Bao TRAN

M2 Master's Student, Grenoble INP, UGA, Grenoble, France

thanh-bao.tran@grenoble-inp.org — +33 (0)7 80 88 61 78 — GitHub — Portfolio

## PROFESSIONAL EXPERIENCE

---

### Robotics Engineer

Apicoo Robotics

Hanoi, Vietnam

04/2024 – 06/2025

- Simulated robotic arms in NVIDIA Isaac Sim for Sim-to-Real applications, including environment modeling and virtual sensor integration.
- Built a 6D object-pose pipeline for robotic grasping: RGB-D and point-cloud acquisition via the Intel RealSense SDK (pyrealsense2), with DBSCAN segmentation + ICP registration (Open3D/PCL) as the classical baseline.
- Adopted learning-based pose estimation: DOPE trained for 4 packaging-type objects spanning distinct shape classes (carton box, beverage can, plastic bottle, jar); FoundationPose and MegaPose evaluated for novel objects.
- Deployed on an industrial arm with an eye-in-hand RealSense camera, solving hand-eye calibration; reached ~85% grasp success over ~100 trials at ~20s per pick cycle.

### Robotics Engineer Intern

CIR Lab, National Taiwan Normal University

Taipei, Taiwan

10/2023 – 04/2024

- Developed a dual-arm bimanual manipulation system (bottle-cap opening) using deep reinforcement learning (PPO) on top of Operational Space Control in NVIDIA Isaac Gym.
- Trained a custom pose CNN (4-channel RGB-D ResNet18) on ~10k domain-randomized images rendered via the Isaac Gym camera API, localizing the bottle and cap to a median error of ~2.2 mm, feeding the policy observation.
- Designed and tuned the observation/action spaces and reward function for the PPO policy commanding 6-DoF delta-poses per arm.
- Closed the full perception-to-policy-to-control loop in simulation and initiated transfer to a real dual Franka Emika setup with a RealSense camera, tackling hand-eye calibration.

### Work-Study Student

RA Lab, Hanoi University of Science and Technology

Hanoi, Vietnam

05/2021 – 09/2023

- Designed and implemented Active Disturbance Rejection Control (ADRC) combined with zero-vibration input shaping for overhead crane systems.
- Conducted both simulation studies and hardware experiments with real-time embedded controllers.

## PROJECTS

---

### Flatness-Based Trajectory Planning and Control for UAV Slung-Load Systems

Advanced Research Project

Grenoble, France

09/2025 – 02/2026

- Designed a cascade control architecture for a quadrotor slung-load system: flatness-based trajectory planning, position ADRC with a known-unknown decomposition of the cable force (nominal tension as feedforward, ESO rejecting only deviations), and geometric attitude control on SO(3).
- Implemented a 10-state Extended Kalman Filter fusing position, cable-angle, and IMU measurements to estimate the full state and cable tension; IMU fusion reduced velocity-estimation RMSE by ~44% over position-only sensing.
- Validated in MATLAB simulation under realistic sensor noise across point-to-point, circular, and figure-8 trajectories.

### Following Me: LiDAR-Based Person Following on a Mobile Robot

Mobile Robotics Course Project

Grenoble, France

09/2025 – 02/2026

- Built a real-time (10 Hz) detection-tracking-control pipeline in ROS2 (C++) on the RobAIR platform (Hokuyo UST-15LX 2D LiDAR): background-subtraction motion detection, scan clustering, and leg-pair person detection.
- Implemented a nearest-neighbor data-association tracker with an adaptive uncertainty radius and a track-persistence counter, robust to occlusion and sensor noise.
- Designed dual PID control (rotation + translation) with velocity limits (max 0.6 m/s), a 0.35 m safety distance, and closest-obstacle avoidance; commanded the robot over Wi-Fi via distributed ROS2 (DDS).
- Validated on recorded rosbags (9 detection / 5 tracking) and 3 scripted failure scenarios; diagnosed and fixed a phantom-person detection bug with a leg-exclusivity constraint.

### Learning Model Predictive Control for Autonomous Racing

L-MPC Coursework Project

Grenoble, France

09/2025 – 02/2026

- Implemented a Learning-MPC framework for autonomous racing on the Barcelona Circuit.
- Combined model predictive control with iterative learning and safe-set-based value function approximation for optimal lap-time performance.

## EDUCATION

---

### Master of Science in Mobile, Autonomous and Robotic Systems

Grenoble INP — University of Grenoble Alpes, Grenoble, France

GPA: 14.62/20.00

09/2025 – 07/2026

Relevant Coursework: Learning MPC (Optimal Control, MPC, Reinforcement Learning), Mobile Robotics (Perception, UAV Control, State Estimation), Autonomous Systems (Path Planning, Multi-Robot Cooperation).

### Engineer's Degree in Control Engineering and Automation

Hanoi University of Science and Technology, Hanoi, Vietnam

GPA: 3.05/4.00

08/2018 – 09/2023

Relevant Coursework: Control Theory, Robotics, Nonlinear Systems, Embedded Systems.

### Exchange Program in Electrical Engineering and Information Technology

Technical University of Munich, Munich, Germany

Erasmus+ Scholar

10/2021 – 04/2022

Relevant Coursework: Advanced Robot Control and Learning, Power Electronics.

## PUBLICATIONS

---

- **Tran Thanh Bao**, Nguyen Minh Duc, Minh Duc Duong, and Trong-Hieu Do, “ZV Shaper–ADRC Combination Control for Crane Systems with Constrained Control Signal”, *Mathematical Modeling and Computing in Automation*, vol. 4, no. 3, pp. 32–38, Dec. 2023.

## HONORS AND AWARDS

---

### Academic Excellence Scholarship

Awarded by Hanoi University of Science and Technology for outstanding academic performance.

Hanoi, Vietnam

02/2023

### TUMexchange Scholarship (Erasmus+)

Highly competitive EU scholarship for academic excellence (1 of 4 students selected in Vietnam).

Munich, Germany

07/2021

## TECHNICAL SKILLS

---

- **Programming:** Python, C++, MATLAB/Simulink
- **Robotics Software:** ROS2, NVIDIA Isaac Gym/Sim, Gazebo, RViz, Autoware (basic)
- **ML & CV:** PyTorch, OpenCV, Open3D/PCL, scikit-learn, Deep RL (PPO), 6D pose estimation (DOPE, MegaPose, FoundationPose)
- **Control & Optimization:** MPC, LQR, ADRC, Optimal Control, Convex Optimization, EKF
- **Hardware Integration:** UAV flight controllers, IMU sensors, depth cameras (Intel RealSense SDK / pyrealsense2)
- **Development Tools:** Git, Linux, Docker, Blender
- **Languages:** Vietnamese (native) — English (fluent, C1) — German (B1) — French (A1)